

Hypothesis Testing with Randomization

IMS Chapter 11 (with a probability & distributions review)

Quick review: probability rules

Probability rules, revisited

A fast refresher before we build on these ideas today.

- ▶ **Complement rule:** $P(A^c) = 1 - P(A)$ — the probability an event does *not* happen.
- ▶ **General addition rule:** $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ — subtract the overlap so it isn't double-counted.
- ▶ **Independence:** A and B are independent if $P(A \cap B) = P(A) \times P(B)$ — equivalently, $P(A | B) = P(A)$.

Conditional probability & Bayes' theorem

► **Conditional probability:**

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

► **Bayes' theorem** flips the direction of a conditional probability — useful when you know $P(A \mid B)$ but want $P(B \mid A)$:

$$P(B \mid A) = \frac{P(A \mid B) P(B)}{P(A \mid B) P(B) + P(A \mid B^c) P(B^c)}$$

We'll lean on **independence** heavily today — the null hypothesis in both of today's case studies is a claim that two variables are independent.

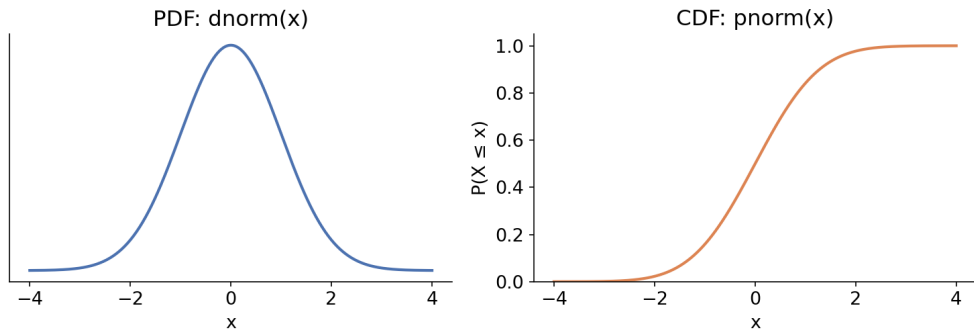
PDF vs. CDF

Recall from last time: a **density curve** describes how probability is distributed across values.

- ▶ **PDF (probability density function):** $f(x)$. In the discrete case, $f(x) = P(X = x)$ exactly. In the continuous case, $f(x)$ is a *density* — the probability of any single exact value is 0, so we only get probabilities from areas.
- ▶ **CDF (cumulative distribution function):** $F(x) = P(X \leq x)$ — the running total of probability *up to* x . This works the same way for both discrete and continuous random variables.

The normal distribution: PDF vs. CDF

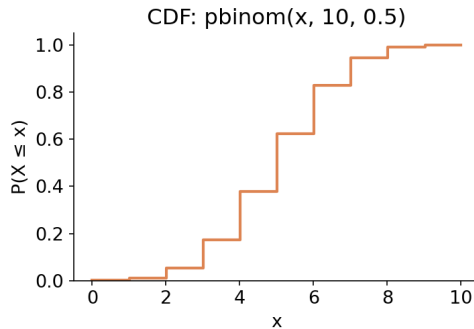
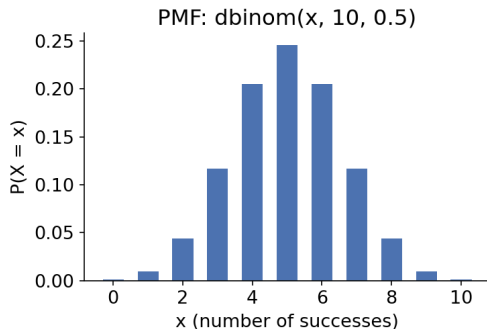
Normal distribution: density vs. cumulative probability



In R: `dnorm(x)` gives the height of the density curve at x ; `pnorm(x)` gives the area to the left of x — i.e., $P(X \leq x)$.

The binomial distribution: PMF vs. CDF

Binomial distribution: PMF vs. cumulative probability



Same idea, discrete version: `dbinom(x, n, p)` gives $P(X = x)$ exactly; `pbinom(x, n, p)` gives $P(X \leq x)$, the running total.

Hypothesis testing with randomization

Where does variability come from?

Suppose your instructor splits the class into students sitting on the **left** vs. **right** side of the room, and asks who prefers reading books on a screen.

If $\hat{p}_L$ and $\hat{p}_R$ are the proportions who prefer screens on each side, would you be surprised if $\hat{p}_L \neq \hat{p}_R$ exactly?

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If $\hat{p}_L$ and $\hat{p}_R$ are the proportions who prefer screens on each side, would you be surprised if $\hat{p}_L \neq \hat{p}_R$ exactly?

Almost certainly not — we'd expect some difference just from **chance**, even if seating side has *nothing* to do with reading preference. The hard question in statistics is: **how do we tell a “real” difference apart from one that’s just chance?**

Case study 1: Sex discrimination

Research question: Are individuals who identify as female discriminated against in promotion decisions?

48 male bank supervisors were each given an identical personnel file and asked whether to promote the candidate — except half the files were randomly labeled “male,” half “female.” (Rosen & Jerdee, 1974)

	Promoted	Not promoted	Total
Male	21	3	24
Female	14	10	24
Total	35	13	48

Promotion rate: **87.5%** (male) vs. **58.3%** (female) — a difference of **29.2 percentage points**.

Setting up the hypotheses

The 29.2% gap is a **point estimate** of a real difference — but the sample is small. Is this discrimination, or could it just be chance?

- ▶ H_0 (**null hypothesis**): sex and decision are independent — the 29.2% gap is due to natural variability.
- ▶ H_A (**alternative hypothesis**): sex and decision are *not* independent — female candidates are discriminated against.

The null hypothesis usually represents a “**no difference**” starting point. We only abandon it (reject!) if the data conflict with it strongly.

The key idea: randomization

If H_0 is true, promotion had nothing to do with the sex on the file — so it wouldn't matter *how* the files got split into “male” and “female” piles.

So let's simulate that. Shuffle all 48 files (35 marked promoted, 13 not) together, and deal them into two new piles of 24 — pretending we relabeled them at random.

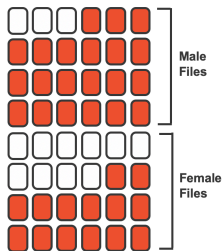
(one simulated shuffle)	Promoted	Not promoted	Total
“Male”	18	6	24
“Female”	17	7	24

Simulated difference: $18/24 - 17/24 = 0.042$ — much smaller than the real 29.2%.

Visualize a Simulated Difference

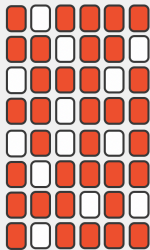
Experiment
 $\hat{p}_M - \hat{p}_F = 0.292$

Gather the Data



Randomization Shuffle #1
 $\hat{p}_{M,shfl1} - \hat{p}_{F,shfl1} = 0.042$

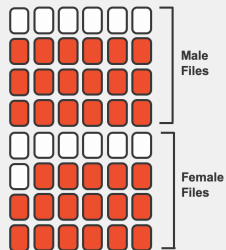
Shuffle



Group

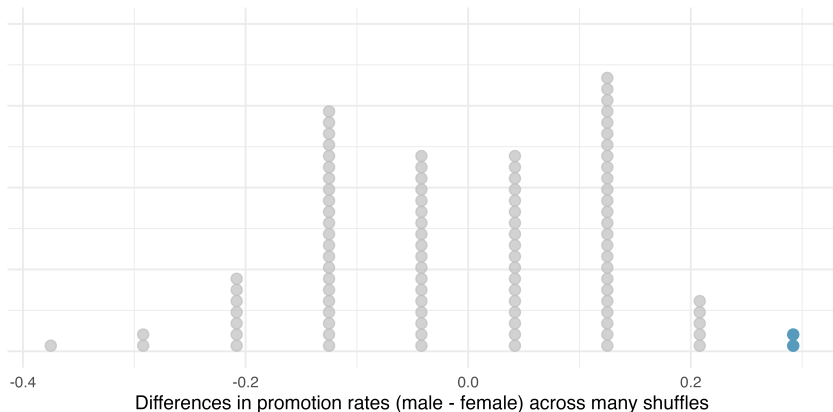


Sort within Group



Repeating the shuffle, thousands of times

One shuffle isn't enough — we need to see the *whole range* of differences chance alone could produce. Repeating this over and over builds a **null distribution**: what the difference in promotion rates looks like when sex truly has no effect.



From simulation to conclusion

Out of thousands of simulated (fake) reshuffles, only about **2.4%** produced a difference as extreme as the real 29.2%.

- ▶ If H_0 is true: we just observed something that would only happen about 2 times in 100 — pretty rare.
- ▶ If H_A is true: female candidates are discriminated against, which is exactly why we'd see such a large gap.

Since a gap this large is rare under H_0 , we **reject the null hypothesis** — the data provide strong evidence of discrimination.

Case study 2: Opportunity cost

Research question: Does reminding college students that unspent money can be saved for later purchases reduce impulse buying?

150 students were shown a \$14.99 video and asked whether they'd buy it. The **treatment** group's option added a reminder: "*...Not buy this video. Keep the \$14.99 for other purchases.*"

	Buy	Not buy	Total
Control	56	19	75
Treatment	41	34	75

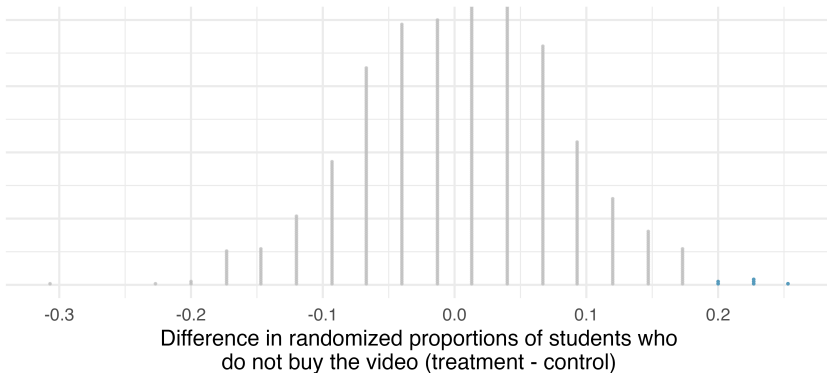
Not-buy rate: **45.3%** (treatment) vs. **25.3%** (control) — a **20 percentage point** difference.

Randomizing the opportunity cost study

H_0 : the reminder has no effect on purchasing decisions. H_A : the reminder reduces the chance of buying.

Same idea: shuffle the 150 decisions (97 buy, 53 not-buy) and re-deal into two groups of 75, pretending group assignment was unrelated to the decision.

1,000 differences in randomized proportions



Conclusion: opportunity cost

Because a 20-point gap is extremely rare under H_0 (roughly 1 in 100 times, if not less), we **reject the null hypothesis**.

Since this was a randomized **experiment** (not just an observational study), we can make a causal claim: reminding students about opportunity cost *causes* a reduction in impulse purchases.

Formalizing: the hypothesis testing framework

- ▶ **Null hypothesis (H_0):** a skeptical claim of “no difference” or “no effect” — assumed true until the data convince us otherwise.
- ▶ **Alternative hypothesis (H_A):** the claim researchers usually set out to find support for — a real relationship or effect.

The US court system analogy: a defendant is presumed innocent (H_0) until evidence convinces the jury of guilt (H_A) beyond reasonable doubt. A “not guilty” verdict doesn’t prove innocence — it just means the evidence wasn’t convincing enough. Same with hypothesis testing: *failing to reject H_0 is not the same as proving H_0 true.*

The p-value

The **test statistic** is the summary value we compute from the data (here, a difference in proportions) and track across simulations.

i p-value

The probability of observing data at least as extreme as ours, **if the null hypothesis were true**.

- ▶ Sex discrimination: $p\text{-value} \approx 0.02$
- ▶ Opportunity cost: $p\text{-value} \approx 0.006\text{--}0.009$

A small p-value means the observed data would be surprising under H_0 — which is evidence *against* H_0 .

Statistical discernibility (a.k.a. “significance”)

We reject H_0 when the p-value falls below a pre-chosen threshold, α (the **discernibility** or **significance level**) — conventionally $\alpha = 0.05$.

- ▶ p-value $< \alpha \rightarrow$ **statistically discernible** evidence against $H_0 \rightarrow$ reject H_0 .
- ▶ p-value $\geq \alpha \rightarrow$ not enough evidence to reject H_0 (this does **not** mean H_0 is proven true).

Careful: “statistically discernible” is a technical statement about the p-value crossing a threshold — it does **not** by itself tell you whether the effect is large or practically meaningful.

The randomization test recipe

1. **Frame the question as hypotheses** — H_0 (no relationship/no effect) vs. H_A .
2. **Collect data** — an experiment if you want a causal claim; an observational study otherwise.
3. **Model the randomness under H_0** — shuffle/permute to build the null distribution.
4. **Analyze the data** — compute the p-value: how often does chance alone produce something this extreme?
5. **Form a conclusion** — reject or fail to reject H_0 , and state it in plain language.

A peek ahead: doing this in code

We did the “shuffle the cards” step by hand (well, by simulation) today. In R, that loop looks like this:

```
# Sex discrimination: shuffle sex labels, recompute the difference, repeat
diffs <- replicate(1000, {
  shuffled <- sample(sex_discrimination$sex)
  # ... recompute difference in promotion rate for the shuffled groups
})

mean(diffs >= 0.292)    # simulated p-value
```

Confidence intervals with bootstrapping

From testing to estimating

Chapter 11 answered yes/no questions: *is there a real difference?* Today's question is different: *how big is it, actually?*

- ▶ Randomization (Ch. 11) models what happens under a **skeptical null** — good for testing a specific claim.
- ▶ **Bootstrapping** (Ch. 12) instead asks: given the one sample we have, what's a plausible **range** for the true population parameter?

Both are simulation-based. Neither requires the normal-approximation conditions from a few weeks ago.

Case study: the medical consultant

A consultant who assists liver-donor surgery patients claims her clients have a lower complication rate than the national average of 10%. Of her **62** clients, **3** had complications.

$$\hat{p} = \frac{3}{62} \approx 0.048$$

Is this strong evidence her work reduces complications?

Two issues: (1) this is **observational** data — even if her rate is lower, we can't claim her work *caused* it (confounders like patient wealth are possible); (2) $\hat{p}$ is just one estimate — how much would it vary if we could repeat this over and over?

The bootstrap idea

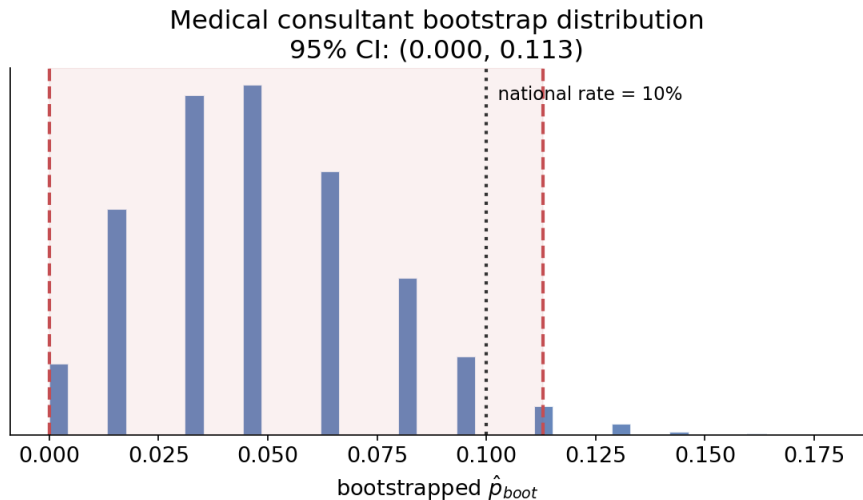
We can't actually go collect thousands of new samples of 62 patients. Instead: treat the **sample itself as a stand-in for the population**, and resample *from it*.

i Bootstrap sample

A sample of the same size as the original, drawn **with replacement** from the original data.

Think of her 62 clients as 62 marbles in a bag (3 red = complication, 59 white = no complication). Draw 62 marbles **with replacement** (so the same marble can be drawn more than once), and compute the resample's proportion, $\hat{p}_{boot}$. Repeat thousands of times.

The bootstrap distribution



The bootstrapped proportions range from 0 to about 0.113. This spread tells us how much $\hat{p}$ could plausibly be wrong and if we could measure

Building the confidence interval

i 95% bootstrap percentile confidence interval

Sort all the bootstrapped $\hat{p}_{boot}$ values. The 95% CI is (2.5th percentile, 97.5th percentile) of that sorted list.

For the medical consultant: **95% CI = (0%, 11.3%).**

Does this interval show her true complication rate is below the national 10% rate?

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Does this interval show her true complication rate is below the national 10% rate?

No — the interval *overlaps* 10%. Her true rate could plausibly be anywhere from 0% up to 11.3%, which includes values both above and below the national average. We can't conclude she's better than average from this data alone.

A sharper example: tappers and listeners

A famous study: 120 “tappers” tap out a well-known tune; 120 “listeners” try to guess the song. Tappers *expected* about 50% of listeners to guess correctly. In reality, only **3 of 120** listeners guessed correctly.

$$\hat{p} = \frac{3}{120} = 0.025$$

Bootstrapping this the same way (marbles: 3 red, 117 white, resampled with replacement, 10,000 times) gives:

95% CI = (0%, 5.83%)

Interpreting a decisive interval

Do the data provide convincing evidence against the tappers' claim that 50% of listeners can guess the tune?

Yes — 50% isn't just outside the interval, it's *far* outside (0% to 5.83%). This is a much more decisive result than the medical consultant case, where the interval straddled the value in question.

The general lesson: whether a confidence interval is useful for testing a specific claim depends on whether that claimed value falls inside or clearly outside the interval — and *how far* outside, if it does.

What is a confidence interval, really?

i Confidence interval

A range of plausible values for an unknown population parameter.

Analogy: a point estimate is like fishing with a spear — aim well, but you'll often miss. A confidence interval is like fishing with a net — cast a wide-enough area and you'll usually catch the fish.

- ▶ Report just $\hat{p}$: probably *not* exactly right.
- ▶ Report a range: good chance of capturing the true p .
Want more certainty of capturing the parameter — do you use a wider or narrower interval?

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Want more certainty of capturing the parameter — do you use a wider or narrower interval?

Wider — a bigger net catches more, but is less precise about exactly where the fish is. This is the tradeoff behind choosing 90% vs. 95% vs. 99% confidence.

The bootstrap process, summarized

1. **Frame the question in terms of a parameter to estimate** — not a yes/no claim this time.
2. **Collect data** — observational or experimental.
3. **Model the randomness by resampling the data itself, with replacement** — the sample stands in for the population.
4. **Build the interval** — pick a confidence level, take the corresponding percentiles of the bootstrap distribution.
5. **Form a conclusion** — state the interval in plain language, and note what it does (and doesn't) tell you.

A peek ahead: bootstrapping in code

```
# Medical consultant: resample WITH replacement, recompute p-hat, repeat
boot_props <- replicate(10000, {
  resample <- sample(consultant_data$outcome, size = 62, replace = TRUE)
  mean(resample == "complication")
})

quantile(boot_props, c(0.025, 0.975)) # 95% bootstrap CI
```

Notice the one-word difference from Chapter 11's code: `replace = TRUE` instead of a plain shuffle. Randomization permutes labels; bootstrapping resamples *with replacement* from the data itself — same simulate-many-times spirit, different question being asked.